

Continuidad de la transformada de Fourier en $\mathcal{L}^1(\mathbb{R})$

Lema 1. Sea $f \in \mathcal{L}^1(\mathbb{R})$, entonces $f^\wedge \in \mathcal{C}(\mathbb{R})$.

Demostración: Sean $\xi, h \in \mathbb{R}$ cualesquiera, entonces

$$f^\wedge(\xi + h) - f^\wedge(\xi) = \int_{\mathbb{R}} f(x) [e^{-2\pi i x(\xi+h)} - e^{-2\pi i x\xi}] dx.$$

Entonces, como el integrando está acotado por $2|f(x)| \in \mathcal{L}^1(\mathbb{R})$, por el Teorema de convergencia dominada, se tiene que

$$\lim_{h \rightarrow 0} [f^\wedge(\xi + h) - f^\wedge(\xi)] = \int_{\mathbb{R}} f(x) \lim_{h \rightarrow 0} [e^{-2\pi i x(\xi+h)} - e^{-2\pi i x\xi}] dx = 0. \quad \blacksquare$$

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